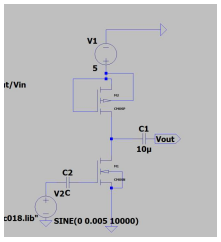
# IA3 & IA4

1

Design and simulate a common source amplifier with diode connected PMOS as load and obtain the following

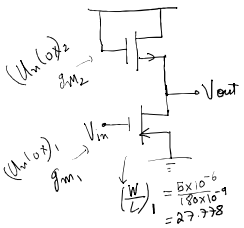
1. A gain (Av) of ____ (last three digits of your Reg. No).
2. Obtain the DC bias voltage needed to operate the transistor in saturation point. Further, explain the method that you adopted for the same
3. Plot the DC and transient characteristics of the CS amplifier

Note : Consider Vdd = 5V, W= 5μM. Vin = $5\times10^{-3}$ sin(20000πt)




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$A_V = \frac{g_{m_1}}{g_{m_2}} = 318$

$\frac{\sqrt{2(\mu_n C_{ox})(\frac{W}{L})_1 I_D}}{\sqrt{2(\mu_n C_{ox})_2 (\frac{W}{L})_2 I_D}} = 318$

from the provided file

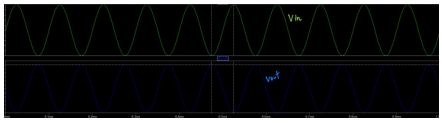
$(\mu_n C_{ox})_1 = 273.8$

$(\mu_n C_{ox})_2 = 115.69$

$\Rightarrow 318^2 = \frac{273.8}{115.69} \times \frac{(W/L)_1}{(W/L)_2} \Rightarrow \frac{(W/L)_1}{(W/L)_2} = 42728.029$

$\left(\frac{W}{L}\right)_2 = 6.5\times10^{-4}$

$W_2 = 0.117\ mm$




b) DC bias Voltage

Assume

$\mu_n C_{ox} = 100\ \mu A/V^2$

$\mu_n C_{ox} = 100\ \mu A/V^2$

$V_{thn} = 0.5$

$V_{thp} = 0.5$

$L_n = L_p = 180nm$

$W_n = 5\mu m$

$W_p = 0.117\ mm$

$I_D = 240\ \mu A$

$\left(\frac{W}{L}\right)_n = 27.78, R_n = 1.6\ A/V^2$

$\left(\frac{W}{L}\right)_p = 42728.029, R_p = 2.77\times10^{-3}\ A/V^2$

$V_{OV,n} = \sqrt{\frac{2I_D}{R_n}} = 0.0173V$

$V_{OV,p} = \sqrt{\frac{2I_D}{R_p}} = 0.415V$

$g_{m_1} = \frac{2I_D}{V_{OV,n}} = 27.7\ mS$

$g_{m_2} = \frac{2I_D}{V_{OV,p}} = 1.157\ mS$

$V_{SG_1p} = |V_{thp}| + V_{OVp} = 0.5 + 0.415 = 0.915V$

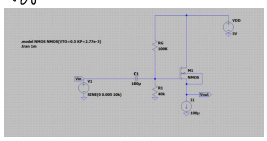
$V_{out,DC} = V_{DD} - V_{SG_1p} = 5 - 0.915 = 4.085\ V$

2

a. Explain the importance of source follower with ideal current source.
b. Design and simulate a source follower with ideal current source for Vdd = 5V, W= 5µM. Vin = $5\times10^{-3}$ sin(20000πt).
c. Plot the transient waveform and comment on the i) gain ii) input impedance and iii) output impedance

a) The Source follower provides a low output impedance buffer (useful to drive loads) and a near unity voltage gain.
Using an ideal current source as the drain bias gives gives a high DC capacitance & a very high small signal output impedance at the drain which helps keeps the transistor in saturation over a wide source swing & improves linearity.

b) $V_{dd} = 5V$, $W = 5um$, $V_{in} = 5\times10^{-3}\sin(2000\pi t)$



Assume $V_{tn} = 0.5V$
$\mu_n C_{ox} = 100\mu A/V^2$
$L = 0.18n$
$I_D = 100\mu A$

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$K_n = \mu_n C_{ox} W/L$
$= 2.778\times10^{-3} A/V$

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{tn})^2$$

$$100\times10^{-6} = \frac{1}{2}\times100\times10^{-6}\times\frac{5}{0.18}(V_{GS}-0.5)^2$$

$$V_{GS} = 0.26 + 0.5$$

$$\boxed{V_{GS} = 0.76}$$

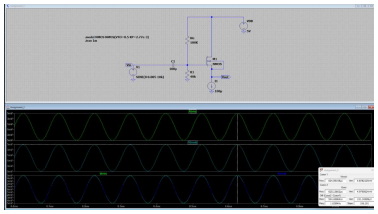
Saturation check
$V_{GS} - V_{tn} \le V_{DS}$
$V_{DS} = V_{DD} = 5V$
$0.76 - 0.5 \le 5$
$\boxed{V_{OV} = 0.26}$

$g_m = 2I/V_{ov}$
$= 745uS$

$A_V = \frac{g_m r_o}{1+g_m r_o} \approx 0.99$

$Z_{in} = R_S = 100k\Omega$
$R_{out} = 1/g_m = 1.34k\Omega$

c)

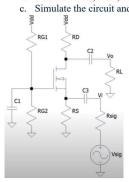


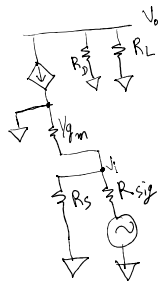
i) Gain = 0.99 (almost 1) in phase with negligible distortion
ii) Input Impedance: Very high (~100 MΩ) as MOS gate draws negligible current
iii) Output Impedance: Low so output closely follows input

3

A common gate amplifier needs to amplify an input of Vin = $1\times10^{-3}$ sin(20000πt) which is been loaded with a 22k Ω load resistor(refer figure). Assume a 100μA current flowing in the bias resistors, VDD = 12V, Zin = 50 Ω.

a. Design: RD, RS, RG1 and RG2
b. Calculate: Zin, Zout, and AV. Assume Rsig = 50 Ω
c. Simulate the circuit and plot transient waveform for Vin and Vout




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→ $Z_{in} = 50\,\Omega$
→ Assume 100μA current

$Z_{in} = R_S \| \frac{1}{g_m}$

$R_S >> \frac{1}{g_m}$

$Z_{in} \approx \frac{1}{g_m}$

With $r_o \to \infty$

$Z_{out} = R_D$

$A_{V_o} = g_m R_D$

$g_m = \sqrt{2 K_n I_D}$

$V_{DS} = I_D R_D = I_D R_S$

$I_D = \frac{1}{2} K_n (V_{GS} - V_{tn})^2$

$V_G = I_D R_S + V_{GS}$

$V_G = \frac{V_{DD} R_{G_2}}{R_{G_2} + R_{G_1}}$

$R_{G_2} + R_{G_1} = \frac{V_{DD}}{I}$

$Z_{in} = \frac{1}{g_m} = 50\,\Omega$

$g_m = \sqrt{2 K_n I_D} = 20\,mA/V$

$I_D = 200\,\mu A$

$\frac{V_{DD}}{3} = I_D R_D = I_D R_S = 4\,V$

$R_D = R_S = 20\,k\Omega$

$\Rightarrow 18\,k\Omega \; E12$

$Z_{out} = R_D = 18\,k\Omega \; E12$

$I_D = \frac{1}{2} K_n (V_{GS} - V_{tn})^2$

$200\mu = \frac{1}{2}(V_{GS} - 0.5)^2$

$\Rightarrow V_{GS} = 0.52\,V$

$V_S = I_D R_S = 3.6\,V$

$V_G = V_S + V_{GS} = 4.12\,V$

$V_G = \frac{V_{DD} R_{G_2}}{R_{G_2} + R_{G_1}}$

$4.12 = \frac{12 R_{G_2}}{R_{G_2} + R_{G_1}}$

$\Rightarrow R_{G_2} + R_{G_1} = \frac{V_{DD}}{I} = 120\,k\Omega$

$R_{G_2} = 41.2k \Rightarrow 39k\,\Omega \; E12$

$R_{G_1} = 81k \Rightarrow 82k\,\Omega \; E12$

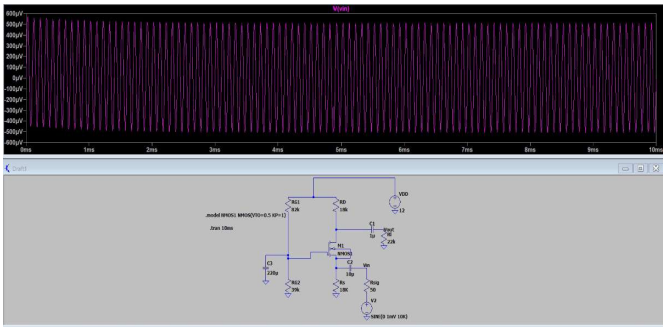
$|A_{V_o}| = g_m R_D = 360\,V/V \Rightarrow 51.1\,dB$

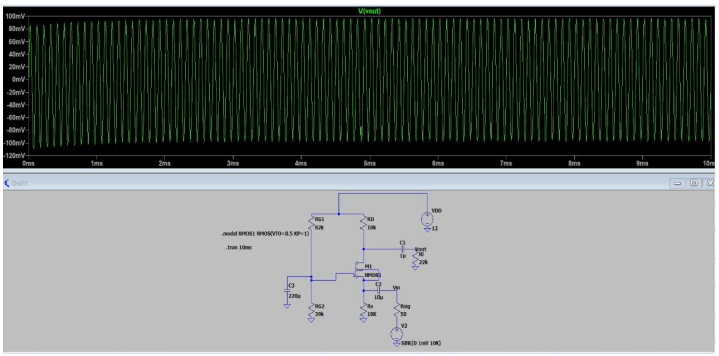
$R_L' = R_D \| R_L = 9.9k\,\Omega$

$|A_V| = g_m R_L' = 198\,V/V \Rightarrow 45.9\,dB$

$G_V = A_V \left(\frac{Z_{in}}{Z_{in} + R_{sig}}\right)$

$G_V = 99\,V/V \Rightarrow 39.9\,dB$





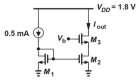

4

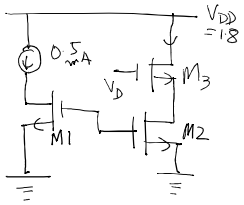
Figure depicts a cascode current source whose value is defined by the mirror arrangement, M1-M2. Assume W/L = 5 μm/ 0.18 μm for M1-M3.

a. Select the value of Vb so that Iout is precisely equal to 0.5 mA.

b. Determine the change in Iout if Vb varies by ±100 mV. Explain the cause of this change.

c. Using hand calculation, determine the output impedance of the cascode section.




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a) $I_{out} = 0.5\,mA$

$I_{out} = \frac{1}{2}\mu_n C_{ox}(V_{GS}-V_{th})^2$

Let $\mu_n C_{ox}\frac{W_1}{L_1} = \mu_n C_{ox}\frac{W_2}{L_2} = \mu_n C_{ox}\frac{W_3}{L_3}$

$= 2.773\times10^{-3}\left[\frac{W_1}{L_1} = \frac{5}{0.18}\right]$

$0.5m = \frac{1}{2}\times 2.773\times10^{-3}(V_{G1}-V_{S1}-V_{th})^2$

$\frac{1}{2.773} = (V_{G1}-0.5)^2$

$0.6005 = V_{G1}-0.5$

$V_{G1} = 1.1005\,V$

$V_{G1} = V_{G2} = 1.1005\,V$

$I_a = \frac{1}{2}\times2.773\times10^{-3}(V_{G2}-V_{S2}-0.5)^2$

$= \frac{1}{2}\times 2.773\times10^{-3}\times(V_b - V_{d2}-0.5)^2$

$V_{G2} = V_b - V_{d2}$

$1.1005 = V_b - V_{d2}$ — (1)

$1.1005 - 0.5 < V_{d2}$

$V_{d2} \geq 0.6005$ — (2)

(2) ~ (1) $\Rightarrow V_b \geq 1.7005$

$V_D \geq 1.7005$

$I_{out} = 0.5\,mA$

(b) $V_6 = 1.7005 + 100mV$

there can't be any change in $I_{out}$ as $I_{out}$ = saturation
$= 0.5mA$

for $V_6 > 1.7005$

$V_{o1} > V_d$ is not possible

as $V_6 \downarrow$ $V_{d2} \downarrow$ $I_{out} \downarrow$

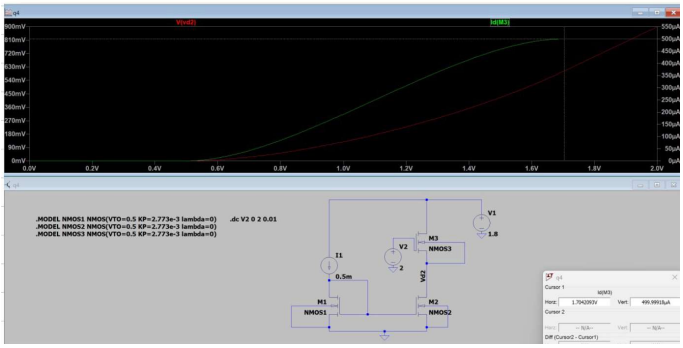
(c) $R_{out} = g_{m3} r_{o3} r_{o2}$

$r_{o3} = r_{o2} = \infty$

So $R_{out} = \infty$

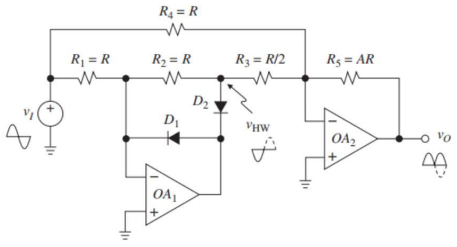
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5

Suitably modify the FWR of Fig. shown, so that, when fed with a triangular wave of ±5V peak values, it gives a triangular wave of ±5V peak values, but twice the frequency. Assume ±10V regulated power supplies.




Without modification it acts as an inverted HWR at point A.

When we add 2.5 V with 10 KΩ in series opamp acts as summing amplifier of $V_{in}$, $V_{HW}$ & 2.5 V

$$V_{out} = -\left(\frac{V_{in}\,20K}{10K} + \frac{V_A \times 20K}{5K} + 2.5 \times \frac{20K}{10K}\right)$$

$$= -(2V_{in} + 4V_A + 5)$$

Here $V_A$ is inverted half wave rectified so its always -ve

$V_{out}$ proof for graph

1) $V_{in} = +2.5$ then $V_A = -2.5$

$V_{out} = -(2(2.5(4\times2.5)+5)$

$= 0$

2) $V_{in} = -2.5$ then $V_A = 0$

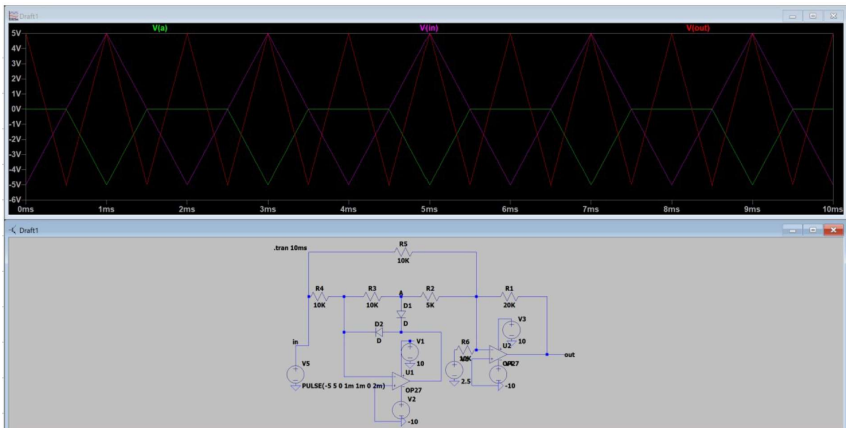
$V_{out} = -(2\times2.5 + 5) = 0$

3) $V_{in} = 5$ then $V_A = -5$

$V_{out} = -(2\times5 + (4\times-5) + 5)$

$= -10 + 20 - 5$

$= 5$

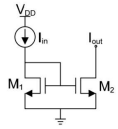


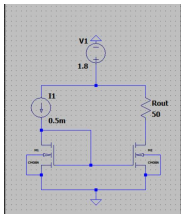

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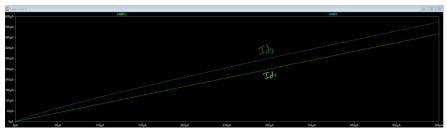
6

Simulate the following circuit (with suitable Rout) and plot $I_{in}$ Versus $I_{out}$ when W/L ratio of a) M1=M2 b) M1=2M2 c) M1=M2/2. Assume Vdd = 1.8V, W/L = 5 µm/ 0.18 µm, $I_{in}$ = 0.5mA




(1)



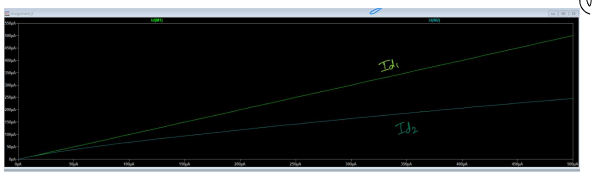

given $V_{DD} = 1.8$

$\frac{W}{L}_{initial} = \frac{5}{0.18}$, $I_{in} = 0.5$ mA

$I_D = \frac{1}{2} K' \left(\frac{W}{L}\right)(V_{GS} - V_{th})^2$

$\Rightarrow \frac{I_{out}}{I_{in}} = \frac{(W/L)_2}{(W/L)_1}$

(2)




(3)

